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Problem Statement

This research develops an optimisation framework for aerial firefighting through spatial discretisation of the operational area into a finite grid, enabling coordinated deployment of a heterogeneous aircraft fleet across multiple water sources. Fire propagation and linear flight dynamics are incorporated to simulate suppression operations, with the objective of minimising fire spread and minimising time to extinguishment.

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Methodology

The methodology is structured into the following components:

2.0.1. Discretisation

The operational area, encompassing the fireline, water sources, and aircraft depots, is discretised into a uniform grid of rectangular cells of fixed size. Each cell is treated as a single point, reducing the problem from a continuous spatial domain to a finite set of discrete elements. This significantly reduces computational complexity while retaining sufficient spatial resolution for meaningful analysis.

A cell can have 3 following states:

$$S = \{0, 1, 2\}$$

where

- 0 refers to unburned cell
- 1 refers to a burning cell
- 2 refers to an extinguished cell and cannot be reignited

A binary decision variable y can be defined as:

$$y_{c,t,s} \in \{0, 1\}$$

which is 1 if cell c at time t has a state s $\in S$

$$\sum_{s \in S} y_{c,t,s} = 1 \forall c \in C, t \in T$$

t itself is discrete and belongs to T at a certain timestamps.

Possible discretisation size can be dictated by the standard deviation of the water drop pattern

2.0.2. Fire Modelling

A cell is only considered burning, if more than 50% of the cell is covered by the burning fireline. Coupling fire propagation with real-time firefighting presents a significant modelling challenge. Two approaches are considered:

Pre-computed Fire Propagation

In this approach, fire propagation is simulated in advance without accounting for any firefighting intervention. The resulting fireline is updated at fixed time intervals within the optimisation. A correction is applied by removing propagation originating from cells that have been extinguished. While computationally efficient, this approach oversimplifies the interaction between suppression efforts and fire spread.

Iterative Fire and Optimisation Coupling
Alternatively, the fireline is held constant over a short time window during which the optimisation is performed. Once the window ends, fire propagation is simulated for the same duration and the updated fireline is passed back to the optimiser. This process repeats iteratively. The main limitation is that abrupt changes in the fireline between iterations may disrupt the continuity of the optimised strategy.

2.0.3. Fire Drop Pattern

The shape of the water dispersion pattern is referenced and is simplified into a hexagonal shape, as indicated by Figure 2.1

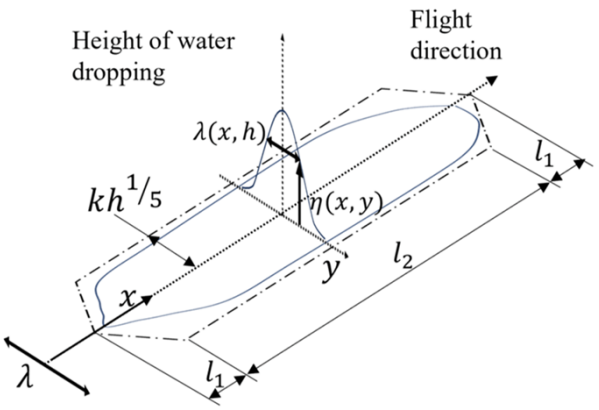


Fig. 4. Schematic of water dispersion pattern.

Figure 2.1: Schematic of water dispersion pattern

The distribution is approximated by a normal distribution. The width is related to the aircraft's speed and descent velocity of the fire retardant. λ is constructed as:

$$\lambda(x,h) = \begin{cases} kh^{1.5}\dfrac{x}{l_1}, & x < l_1, \\ kh^{1.5}, & l_1 \le x \le l_1 + l_2, \\ kh^{1.5}\dfrac{2l_1 + l_2 - x}{l_1}, & l_1 + l_2 < x \le 2l_1 + l_2. \end{cases}$$

$$\eta(x,y) = \frac{\lambda}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{y^2}{2\sigma^2}\right)$$

k serves as the scaling factor to determine the maximum width of the water discharge zone at the current release altitude. l_1 and l_2 characterize the longitudinal distribution. Standard Deviation $\sigma = \lambda/6$.

Water release height are generally around heights between 10m to 80m.

So if $x^a_{jk} = 1$ and $t_k \ge t_j + k\Delta T - M(1 - x^a_{jk})$ then the fire intensity at k is decreased by ϕ and the neighbouring cells is decreased by $f(\phi, d)$ where d is the distance to the cell.

2.0.4. Model Scope

The following physical phenomena and operational elements are modelled:

- Fire propagation
- Wind field (stochastically generated)
- Water drop pattern

- Linear flight dynamics
- Aircraft routing subject to water capacity constraints (variant of the Travelling Salesman Problem)

2.0.5. Assumptions

The following assumptions are applied to make the problem tractable:

- Each grid cell is treated as a discrete unit with one of three states: unburned, burning, or extinguished.
- Fire propagation follows the behaviour described by the selected fire spread model.
- Water drop patterns are modelled as a normal distribution over the affected cells.
- No unexpected events such as aircraft failure or sudden fireline discontinuities are considered.
- Once a cell is extinguished, reignition is not permitted.
- Wind velocity is sampled from a prescribed stochastic profile and is assumed uniform within each cell. The same wind profile is used in both the optimisation and fire propagation simulation.
- Aircraft flight time is assumed independent of water load and airspeed variation.
- Scooping aircraft are assumed to fill their water tanks only when operating at zero ground speed over a water source.
- Flight path continuity is enforced through linear dynamic constraints to prevent physically unrealistic manoeuvres.